Week 5 - Learning Map

Goal:

Understand multimodal conversational system design and apply LangGraph concepts like subgraphs, persistence, and HITL to build a stateful multi-agent e-commerce assistant that supports voice interaction, intelligent routing, semantic product search, support workflows, and unified customer responses.

Overview...

This module introduces multimodal conversational systems and shows how to turn those concepts into a practical multi-agent customer support assistant.

You first learn how voice-enabled systems work through the core pipeline of speech-to-text, language reasoning, and text-to-speech, along with the difference between cascaded and speech-to-speech architectures.

The module then explains how LangGraph helps structure these systems using subgraphs, state management, persistence, and memory so workflows can remain modular and stateful across multiple turns.

It also introduces human-in-the-loop design using interrupts, where execution can pause for review or missing information and then resume safely.

These concepts are applied in the AxiomCart AI Assistant project, a multi-agent e-commerce support system that combines an orchestrator, product search agent, support agent, and response synthesizer.

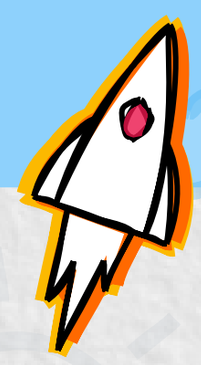
The system supports semantic product search, order tracking, escalation to human support, and conversation memory, while also enabling voice-based interaction.

Together, the theory and project show how multimodal workflows, LangGraph orchestration, and agent collaboration come together in a production-style assistant.

Note:

Post-class resources are optional and intended for post-class learning.

HOW TO START?



Roadmap

Pre-class Setup

- Clone the project repo
- Install required dependencies
- Set up OpenAI API key and environment variables

Learners are encouraged to follow along with the instructor during the live session and execute the code on their local environment.

Live Class

Multimodal System Foundations

- STT → LLM → TTS pipeline
- Cascaded vs speech-to-speech architecture
- How voice-based conversational agents work

LangGraph Workflow Design

- Subgraphs and reusable workflows
- Persistence and memory
- Human-in-the-loop with interrupts

Project: AxiomCart AI Assistant

- Orchestrator + product/support agents
- RAG, tool calling, escalation, memory
- Voice-enabled customer support experience

Assignment

Post-class resources (Optional deep dives)

GenAI for Voice

- Text to Speech, Speaker Diarization
- WaveNet, Voice with LLMs
- Vocoders

Please use the Wednesday Technical Coaching session to clarify any doubts you have on the topics. The assignment solutions will be discussed during the Thursday Assignment Review session.

All the resources have been added to your Uplevel account. If you have any questions, please reach out to the Operations team or raise a support ticket.